

Supersonic Fighters

The 1950s was a time of tremendous change in the fighter world. Ever more powerful engines and an increasing understanding of supersonic aerodynamics, fueled by Cold War paranoia and big research and development budgets, saw top speeds rise from barely breaking the sound barrier in a dive to greater than Mach 2—twice the speed of sound—in level flight.



△ **Convair F-102 A Delta Dagger 1953**
Origin USA
Engine 16,000 lb (7,257 kg) thrust Pratt & Whitney J-57 turbojet
Top speed 824 mph (1,328 km/h)

A bold, innovative design, this tailless delta-wing interceptor was initially a huge disappointment as the prototype was incapable of supersonic flight. The improved F102A was given an area-ruled “coke-bottle” shaped fuselage and reached Mach 1.22 in flight.



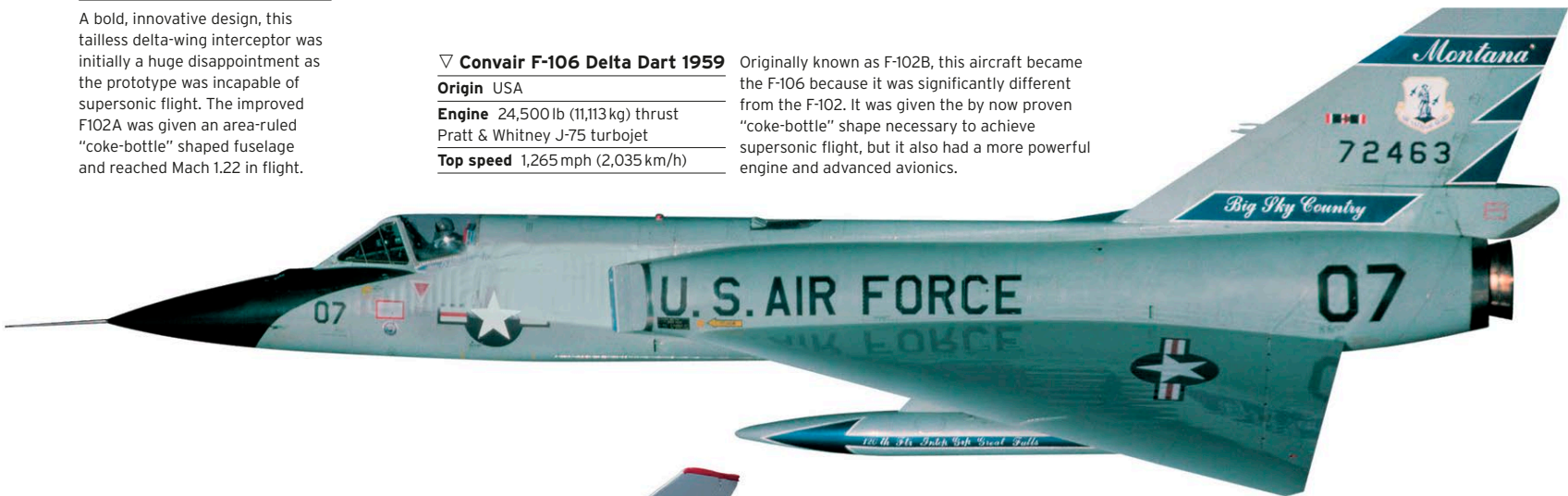
▽ **Dassault MD-452 Mystere IVA 1952**
Origin France
Engine 7,716 lb (3,500 kg) thrust Hispano-Suiza Verdon 350 turbojet
Top speed 695 mph (1,120 km/h)

Although descended from the Mystere II, the IVA fighter-bomber was capable of supersonic flight. Originally powered by a Rolls-Royce Tay turbojet, most production aircraft were fitted with an engine built under license by Hispano-Suiza.



▽ **Convair F-106 Delta Dart 1959**
Origin USA
Engine 24,500 lb (11,113 kg) thrust Pratt & Whitney J-75 turbojet
Top speed 1,265 mph (2,035 km/h)

Originally known as F-102B, this aircraft became the F-106 because it was significantly different from the F-102. It was given the by now proven “coke-bottle” shape necessary to achieve supersonic flight, but it also had a more powerful engine and advanced avionics.



◁ **Grumman F11F-1 Tiger 1956**
Origin USA
Engine 10,500 lb (4,763 kg) thrust Wright J-65 turbojet
Top speed 727 mph (1,170 km/h)

The poor range and endurance of the US Navy’s second supersonic fighter resulted in a short career with the fleet being phased out of operations by 1961. The Tiger was flown by the Navy’s aerobatic team, the Blue Angels, until 1968.



△ **Mikoyan-Gurevich MiG-19 1955**
Origin USSR
Engine 2 x 7,178 lb (3,256 kg) thrust Tumansky RD-9B turbojets
Top speed 909 mph (1,455 km/h)

Known by NATO as the “Farmer,” the MiG-19 was the first Soviet fighter to be capable of sustained supersonic flight. Although around 5,500 were produced, it was not as popular as the MiG-17, which it replaced, or the MiG-21, which superseded it.



△ **Mikoyan-Gurevich MiG-21 1959**
Origin USSR
Engine 12,655 lb (5,740 kg) thrust Tumansky R-11F-300 turbojet
Top speed 1,385 mph (2,230 km/h)

Lighter than its Western contemporaries, the MiG-21 “Fishbed” was designed by the Mikoyan-Gurevich design bureau and nicknamed the “balaika” because the planform view resembled the instrument. It flew at speeds in excess of Mach 2.

